



2018 HSC Course Assessment Task 3
(Trial Examination)
Wednesday June 27, 2018.

- Working time –3 hours (plus 5 minutes reading time).
- Write using blue or black pen. Diagrams may be sketched in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt all questions.

- Mark your answers on the answer sheet provided.

- Commence each new question on a new page.
- Show all necessary working in every question. Marks may be deducted for illegible or incomplete working.

BOOKLETS USED:

☐ Mr Lin

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Section 1: Multiple Choice– 1 mark each.

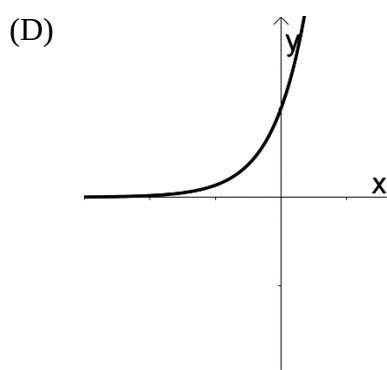
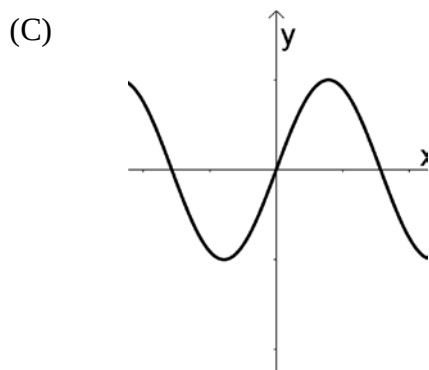
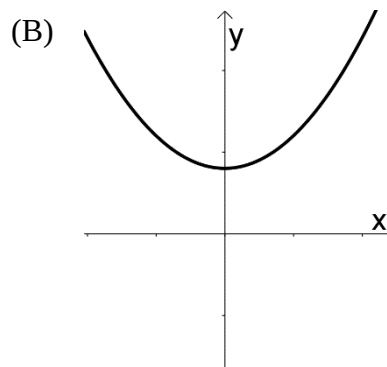
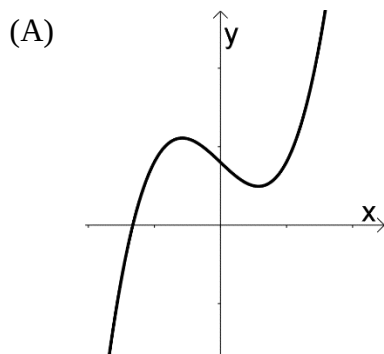
Q1. Which line is perpendicular to the line $4x + 3y + 2 = 0$?

- (A) $4x + 3y - 2 = 0$
- (B) $4x - 3y + 2 = 0$
- (C) $3x + 4y + 2 = 0$
- (D) $3x - 4y - 2 = 0$

Q2. The value of $\sum_{k=2}^{20} 10 - 3k$ is

- (A) -342
- (B) -414
- (C) -437
- (D) -500

Q3. Which function is an odd function?



Q4. Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = x e^x$ between $x = 1$ and $x = 3$?

(A) $\frac{1}{4}(e + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

(B) $\frac{1}{4}(e + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

(C) $\frac{1}{2}(e + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

(D) $\frac{1}{2}(e + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

Q5. What is the period of $y = 3 \tan(4x)$?

(A) $\frac{\pi}{8}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $\frac{2\pi}{3}$

Q6. The solution to $3x^2 + 2x > 8$ is

(A) $-\frac{4}{3} < x < 2$

(B) $x < -\frac{4}{3}, x > 2$

(C) $x < -2, x > \frac{4}{3}$

(D) $-2 < x < \frac{4}{3}$

Q7. If α and β are roots of the equation $2x^2 - 4x - 1 = 0$, what is the value of $\alpha^2 + \beta^2$?

- (A) 3
- (B) 4
- (C) 5
- (D) None of the above

Q8. It is known that $\ln 3a = \ln b - 2 \ln c$, where $a, b, c > 0$.

Which statement is true?

(A) $a = \frac{b - c^2}{3}$

(B) $a = \frac{b}{3c^2}$

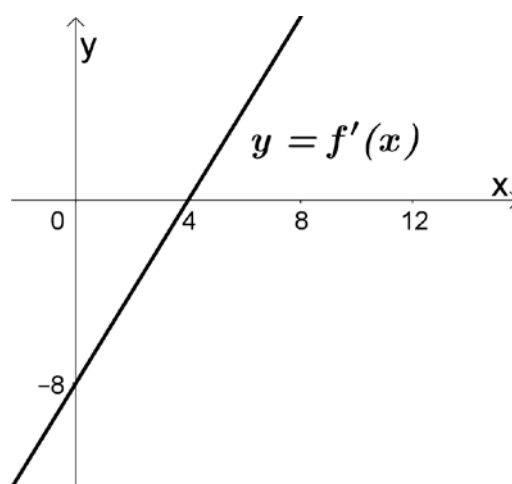
(C) $\ln 3a = \frac{b}{c^2}$

(D) $\ln 3a = \frac{\ln b}{\ln c^2}$

Q9. Evaluate $\int_0^6 |x - 2| dx$

- (A) 10
- (B) 20
- (C) 30
- (D) None of the above.

Q10. The graph of $y = f'(x)$ is shown.



The curve $y = f(x)$ is tangential to the x -axis.

What is the equation of the curve $y = f(x)$?

- (A) $y = 2x^2 - 8x + 8$
- (B) $y = 2x^2 - 8x + 16$
- (C) $y = x^2 - 8x + 8$
- (D) $y = x^2 - 8x + 16$

End of Section I

Section II – Short Answer 90 marks

Question 11 (15 marks) Commence on a NEW page.

Marks

- (a) Rationalise the denominator $\frac{1 - \sqrt{5}}{6 + \sqrt{5}}$ 2
- (b) Factorise fully $16 - 4x^2$. 2
- (c) State the domain of the function $y = \sqrt{4 - x}$. 1
- (d) Solve $|2x - 1| < 4$. 2
- (e) Differentiate $y = 5x^6 - \sqrt{x}$. 2
- (f) Differentiate $y = (\cos x - x)^3$. 2
- (g) Find $\int (3x + 1)^4 dx$. 2
- (h) Solve $\cos\left(\frac{x}{2}\right) = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 4\pi$. 2

End of Question 11

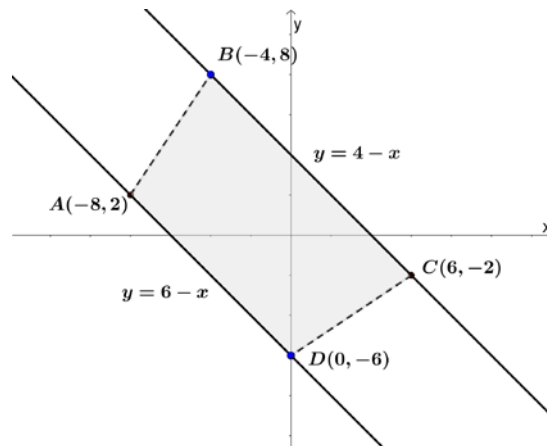
Question 12 (15 marks) Commence on a NEW page.

Marks

- (a) The points $A(-8, 2)$, $B(-4, 8)$, $C(6, -2)$, $D(0, -6)$ define a trapezium in the Cartesian plane.

The equation of the line BC is $y = 4 - x$, and of line AD is $y = 6 - x$.

The distance AD is $8\sqrt{2}$ units.

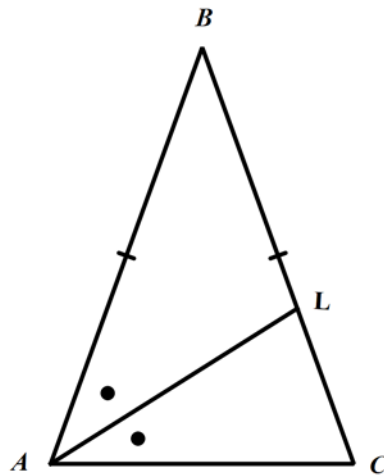


- i) Find the perpendicular distance from the point A to the line BC. 2
- ii) Hence calculate the area of the trapezium. 2
- (b) Differentiate $y = \log_e \frac{3x - 1}{(x + 1)^4}$ 2
- (c) Find:
- i) $\int x(3 - \sqrt{x}) dx$ 2
- ii) $\int \sin(5x + 2) dx$ 2
- iii) $\int \frac{x - 5}{x^2 - 10x} dx$ 2

Question 12 continues on the next page.

Question 12 (continued)

- (d) In a triangle ABC , $AB = BC$. The point L is on BC such that AL bisects $\angle BAC$.



- i) Copy the diagram into your workbook.
- ii) If $AL = AC$, find the size of $\angle ABC$, giving reasons.

3

High quality setting out is required for full marks.

End of Question 12

Question 13 (15 marks) Commence on a NEW page.

Marks

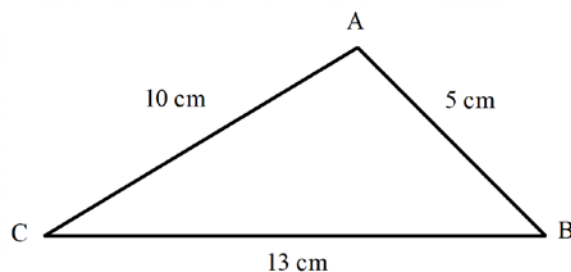
- (a) State the location of the vertex and the focus of the parabola

3

$$8x - y^2 + 6y - 1 = 0.$$

- (b) Triangle ABC has sides $AB = 5$ cm, $BC = 13$ cm and $AC = 10$ cm.

3



Find the exact value of $\tan C$ in simplest form.

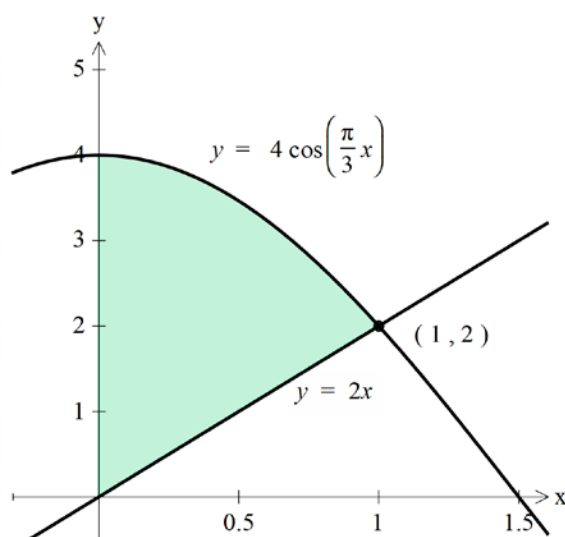
- (c) Sketch the region $y \leq \sqrt{25 - x^2}$ and $y < x$.

3

- (d) Find the values of k for which $y = 5x^2 + (20 - k)x + 20$ is positive definite.

3

- (e) The curve $y = 4 \cos\left(\frac{\pi}{3}x\right)$ meets the line $y = 2x$ at the point $(1, 2)$ as shown in the diagram below.



Find the exact value of the shaded area.

3

End of Question 13

Question 14 (15 marks) Commence on a NEW page.

Marks

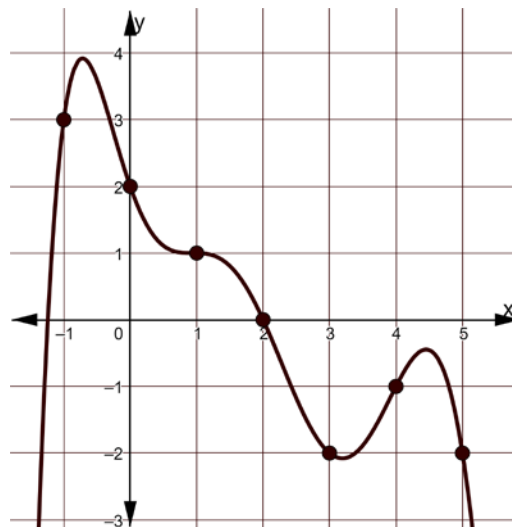
(a) Given

$$f(x) = (x + 2)(x - 2)^3$$

and

$$f'(x) = 4(x - 2)^2(x + 1) = 4x^3 - 12x^2 + 16,$$

- i) Find the stationary points of $y = f(x)$ and determine their nature. 3
 - ii) Find the coordinates of any points of inflexion. 2
 - iii) Sketch the graph of $y = f(x)$, clearly indicating the intercepts, stationary points and points of inflexion. 2
- (b) Given that $x + y$, $x - y$, xy form an arithmetic sequence, write an expression for x in terms of y . 2
- (c) Given the graph of $y = f(x)$ below, 3

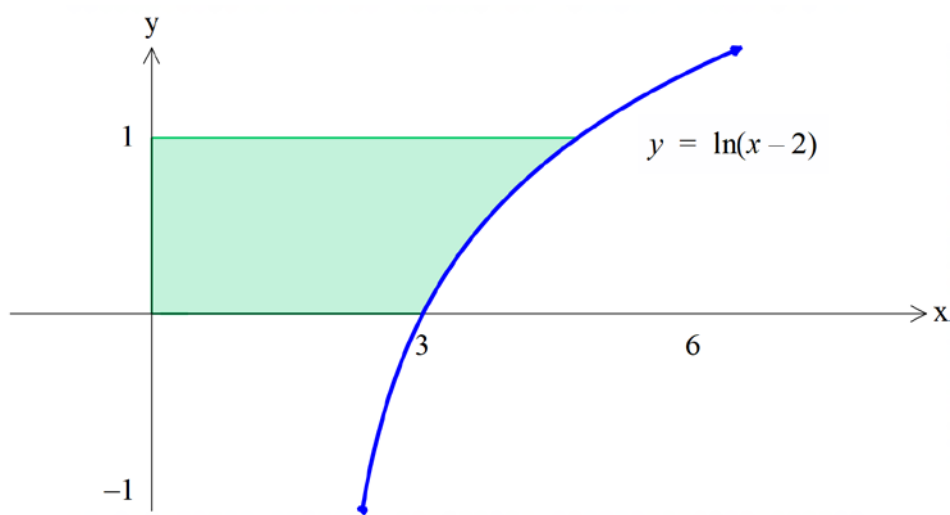


Apply Simpson's Rule with five values to find an approximation for $\int_0^4 f(x) dx$.

Question 14 continues on the next page

- (d) A region is defined by the function $y = \log_e(x - 2)$, the y -axis, and the lines $y = 0$ and $y = 1$. 3

Find the volume of the solid of revolution formed by rotating the region about the y -axis.



End of Question 14

Question 15 (15 marks) Commence on a NEW page.

Marks

- (a) The velocity of a particle travelling along the x -axis is given by the equation

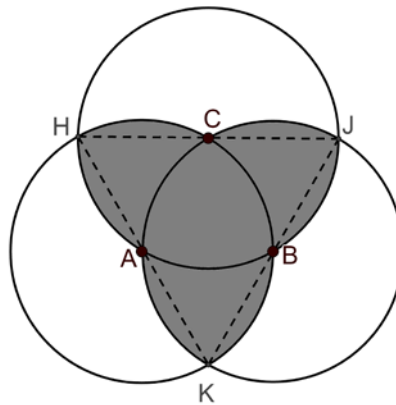
$$v = 3 - \frac{12}{2 + t}$$

where t is the time in seconds and the velocity is in m/s.

- | | | |
|------|---|---|
| i) | When is the particle stationary? | 1 |
| ii) | What happens to the velocity as $t \rightarrow \infty$? | 1 |
| iii) | Sketch the graph of $v(t)$ for $t \geq 0$, showing any intercepts. | 1 |
| iv) | Find the acceleration when the particle is stationary. | 2 |
| v) | Find the distance travelled in the first 6 seconds. | 3 |
-
- | | | | |
|-----|-----|---|---|
| (b) | i) | Show that $\frac{d}{dx} x^2 e^{-x^2} = 2x e^{-x^2} - 2x^3 e^{-x^2}$ | 2 |
| | ii) | Hence find $\int x^3 e^{-x^2} dx$ | 2 |

Question 15 continues on the next page

- (c) Three circles of radius 1 unit with centres A , B and C respectively are arranged as shown in the diagram below.



- i) Find the exact value of the area of the triangle HJK . 1
- ii) Hence or otherwise find the exact value of the shaded area. 2

End of Question 15

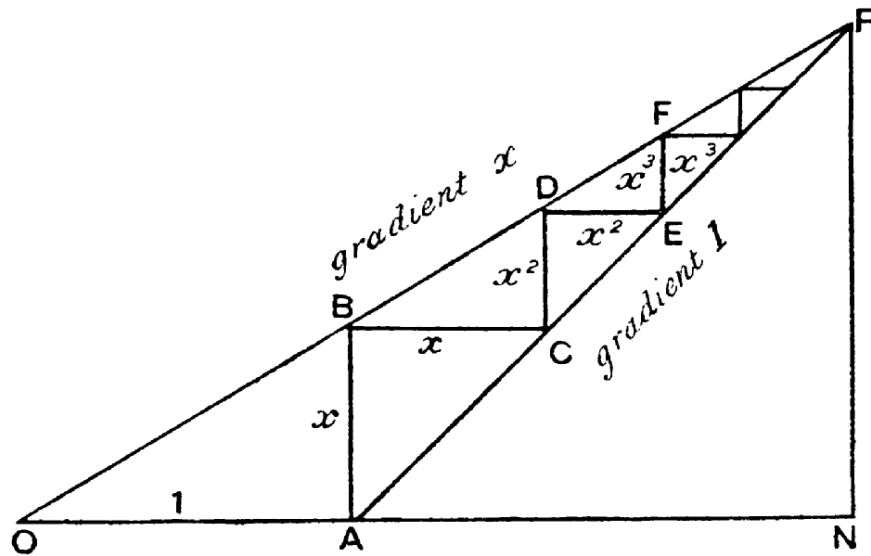
Question 16 (15 marks) Commence on a NEW page.

- (a) By using the property $x = y^{\log_y x}$, or otherwise, show that

2

$$a^{\log_b x} = x^{\log_b a}$$

- (b) The following diagram appears in the 1913 book “Carslaw’s Plane Trigonometry”:



This diagram has become a famous “Proof Without Words” for the sum of an infinite geometric series.

In the questions below, the aim is to *prove* the result for the sum of an infinite geometric series, so don’t use the result in your working out.

We can see from the diagram that $ON = 1 + x + x^2 + x^3 + \dots$.

- i) Explain why $NP = ON - 1$. 1
- ii) Explain why $NP = x ON$. 1
- iii) Using the results (i) and (ii) above, show that, 1

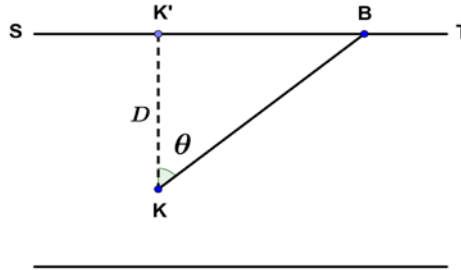
$$1 + x + x^2 + x^3 + \dots = \frac{1}{1-x}$$

- iv) How does this proof demonstrate that x must satisfy the restriction $x < 1$? 1

Question 16 continues on the next page

- (c) At midday, Keanu is in a speedboat on a river at location K when he receives a call from Sandra at location S, riding a bus along a coastal highway towards T.

Sandra asks Keanu to meet up with her bus further along the highway at a location of his choosing – she doesn't mind where they meet, just so long as they eventually meet.



The bus is travelling at a constant speed of V km/hr, scheduled to pass K' at 1 PM. The distance KK' is D km.

Keanu leaves at midday on a bearing of angle θ and meets the bus at point B.

- i) Show that the bus arrives at point B at time t hours,

1

$$t = \frac{D \tan \theta}{V} + 1$$

- ii) Hence show that Keanu will need to travel at a speed r , where

2

$$r = \frac{D V \sec \theta}{D \tan \theta + V}$$

- iii) Show that

2

$$\frac{dr}{d\theta} = \frac{DV \sec \theta (V \tan \theta - D)}{(D \tan \theta + V)^2}$$

given: $\frac{d}{d\theta} \sec \theta = \sec \theta \tan \theta$

- iv) Show that r is minimised when $\tan \theta = \frac{D}{V}$ (reasoning required).

2

- v) Show that the minimum speed r is given by $r = \frac{DV}{\sqrt{D^2 + V^2}}$.

2

END OF THE EXAMINATION.

2018 Mathematics HSC Course Task 3

Suggested Responses

(g) (2 marks)

1. (D) 2. (C) 3. (C) 4. (A) 5. (B)
 6. (C) 7. (C) 8. (B) 9. (A) 10. (D)

Question 11 :

(a) (2 marks)

$$\begin{aligned}\frac{1-\sqrt{5}}{6+\sqrt{5}} &= \frac{(1-\sqrt{5})}{6+\sqrt{5}} \times \frac{(6-\sqrt{5})}{(6-\sqrt{5})} \\ &= \frac{(1-\sqrt{5})(6-\sqrt{5})}{36-5} \\ &= \frac{6-\sqrt{5}-6\sqrt{5}+5}{31} \\ &= \frac{11-7\sqrt{5}}{31}\end{aligned}$$

(b) (2 marks)

$$\begin{aligned}16-4x^2 &= 4(4-x^2) \\ &= 4(2+x)(2-x)\end{aligned}$$

(c) (1 mark)

$$x \leq 4$$

(d) (2 marks)

$$\begin{aligned}-4 &\leq 2x-1 \leq 4 \\ -3 &\leq 2x \leq 5 \\ -\frac{3}{2} &\leq x \leq \frac{5}{2}\end{aligned}$$

(e) (2 marks)

$$\begin{aligned}y &= 5x^6 - x^{\frac{1}{2}} \\ y' &= 30x^5 - \frac{1}{2\sqrt{x}}\end{aligned}$$

(f) (2 marks)

$$\begin{aligned}y &= (\cos x - x)^3 \\ y' &= 3(\cos x - x)^2 \times (-\sin x - 1) \\ &= -3(\sin x + 1)(\cos x - x)^2\end{aligned}$$

$$\begin{aligned}\int (3x+1)^4 dx \\ &= \frac{(3x+1)^5}{5 \times 3} + C \\ &= \frac{(3x+5)^5}{15} + C\end{aligned}$$

(h) (2 marks)

$$\begin{aligned}\cos \frac{x}{2} &= \frac{\sqrt{3}}{2} & 0 \leq x \leq 4\pi \\ & & 0 \leq \frac{x}{2} \leq 2\pi \\ \frac{x}{2} &= \frac{\pi}{6}, \frac{11\pi}{6} \\ \frac{x}{2} &= \frac{2\pi}{6}, \frac{22\pi}{6} \\ x &= \frac{\pi}{3}, \frac{11\pi}{3}\end{aligned}$$

Question 12 :(a) i. (2 marks) BC: $x+y-4=0$ A(-8,2)

$$\begin{aligned}d &= \frac{|1(-8) + 1(2) - 4|}{\sqrt{1^2 + 1^2}} \\ &= \frac{|-8 + 2 - 4|}{\sqrt{2}} \\ &= \frac{10}{\sqrt{2}} = 5\sqrt{2} \text{ units}\end{aligned}$$

ii. (2 marks)

$$\begin{aligned}(BC)^2 &= (-4-6)^2 + (8+2)^2 \\ BC &= 2\sqrt{10}\end{aligned}$$

$$\begin{aligned}\text{Area trapezium} &= 5\sqrt{2} \times \frac{8\sqrt{2} + 10\sqrt{2}}{2} \\ &= 90 \text{ units}^2\end{aligned}$$

(b) (2 marks)

$$\begin{aligned}y &= \ln \frac{3x-1}{(x+1)^4} \\ &= \ln(3x-1) - 4\ln(x+1) \\ y' &= \frac{3}{3x-1} - \frac{4}{x+1}\end{aligned}$$

(c) i. (2 marks)

$$\begin{aligned}
 & \int x(3 - \sqrt{x}) dx \\
 &= \int (3x - x^{\frac{3}{2}}) dx \\
 &= \frac{3}{2}x^2 - \frac{2}{5}x^{\frac{5}{2}} + C
 \end{aligned}$$

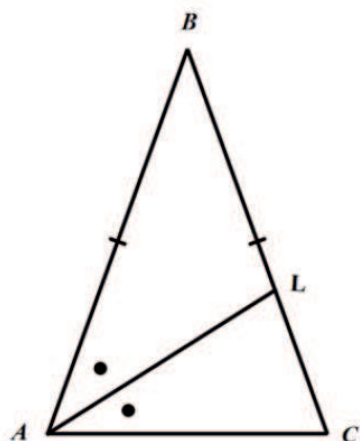
ii. (2 marks)

$$\begin{aligned}
 & \int \sin(5x + 2) dx \\
 &= -\frac{1}{5} \cos(5x + 2) + C
 \end{aligned}$$

iii. (2 marks)

$$\int x^2 - 10x dx = \frac{1}{2} \ln(x^2 - 10x) + C \quad \text{(b) (3 marks)}$$

(d) (2 marks)



Let $\angle BAL = \angle BAC = \theta$ (given)
 $\triangle ABC$ is isosceles (two equal sides)
 $\therefore \angle ACB = \angle BAC = 2\theta$ (base angles of isosceles triangle)

$\triangle ALC$ is isosceles (two equal sides, given)
 $\therefore \angle ACB = \angle ALC$
 $\therefore \angle ALC = 2\theta$

In $\triangle ALC$,
 $\angle LAC + \angle ACL + \angle CLA = 180^\circ = 5\theta$
 (angle sum of a triangle)
 $\therefore \theta = \frac{180}{5} = 36^\circ$

In $\triangle ABC$,
 $\angle ABC + 2\theta + 2\theta = 180^\circ$ (angle sum of a

triangle)

$$\therefore \angle ABC = 180^\circ - 2 \times 36^\circ - 2 \times 36^\circ = 36^\circ$$

Question 13 :

(a) (3 marks)

$$8x - y^2 + 6y - 1 = 0$$

$$y^2 - 6y = 8x - 1$$

$$y^2 - 6y + 9 = 8x - 1 + 9$$

$$(y - 3)^2 = 8(x + 1)$$

(draw a diagram!)

vertex: $(-1, 3)$ focus: $(1, 3)$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos C = \frac{10^2 + 13^2 - 5^2}{2 \times 10 \times 13}$$

$$\cos C = \frac{61}{65}$$

(draw a diagram!)

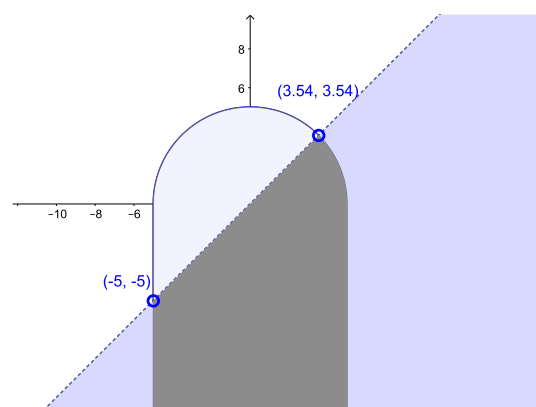
$$x = \sqrt{65^2 - 61^2} = 6\sqrt{14}$$

$$\tan C = \frac{x}{61}$$

$$\tan C = \frac{6\sqrt{14}}{61}$$

(c) (3 marks)

$$y \leq \sqrt{25 - x^2}, y \leq x$$



(d) (3 marks) $y = 5x^2 + (20 - k)x + 20$
 is positive definite when $\Delta < 0$

$$(20 - k)^2 - 4(5)(20) < 0$$

$$400 - 40k + k^2 - 400 < 0$$

$$k^2 - 40k < 0$$

$$k(k - 40) < 0$$

(draw a diagram)

$$0 < k < 40$$

(e) (3 marks)

$$\begin{aligned} \text{Area} &= \int_0^1 (4 \cos \frac{\pi}{3} x - 2x) dx \\ &= \left[-\frac{12}{\pi} \sin \frac{\pi}{3} x - x^2 \right]_0^1 \\ &= (0 - 0) - \left(-\frac{12}{\pi} \sin \frac{\pi}{3} - 1^2 \right) \\ &= \frac{6\sqrt{3} - \pi}{\pi} \text{ units}^2 \end{aligned}$$

Question 14 :

(a) $f(x) = (x + 2)(x - 2)^3$

$$f'(x) = 4(x - 2)^2(x + 1) = 4x^3 - 12x^2 + 16$$

i. (3 marks)

Stationary when $f'(x) = 0$

ie: $4(x - 2)^2(x + 1) = 0$

$$x = 2 \text{ or } x = -1$$

$$f(2) = 0, f(-1) = 27$$

Stationary points at $(2, 0)$ $(-1, 27)$

x	-2	-1	0	2	3
$f'(x)$	-64	0	16	0	16
	\	—	/	—	/

$\therefore$ horizontal point of inflexion at $(2, 0)$, local minimum at $(-1, 27)$

ii. (2 marks)

$$f''(x) = 12x^2 - 24x$$

Possible points of inflection at $f''(x) = 0$ ie: $12x^2 - 24x = 0$

$$x(x - 2) = 0$$

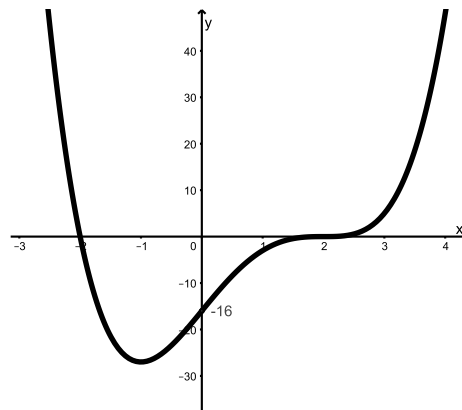
$$x = 0 \text{ or } x = 2$$

We have already shown $x = 0$ is a point of inflexion, test for $x = 2$

x	1	2	3
$f''(x)$	-12	0	36
	$\cap$	—	$\cup$

$\therefore$ Points of inflexion at $(2, 0)$, $(0, -16)$

iii. (2 marks)



(b) (2 marks) AP: $x + y$, $x - y$, xy

$$a = x + y, d = x - y - (x + y) = -2y$$

$$T_3 - T_2 = T_2 - T_1$$

$$xy - (x - y) = (x - y) - (x + y)$$

$$xy - x + y = x - y - x - y$$

$$xy = -2y + x - y$$

$$xy - x = -2y - y$$

$$x(y - 1) = -3y$$

$$x = \frac{3y}{1 - y}$$

(c) (3 marks) Using Simpson's rule:

x	0	1	2	3	4
$f(x)$	2	1	0	-2	-1
	$\times 1$	$\times 4$	$\times 2$	$\times 4$	$\times 1$

$$\int_0^4 f(x) dx \approx \frac{1}{3}(2 + 1 \times 4 + 0 \times 2 + (-2) \times 4 + -1)$$

$$= -1$$

(d) (3 marks) $V = \pi \int_0^1 x^2 dy$

$$y = \ln(x - 2)$$

$$x - 2 = e^y$$

$$x = e^y + 2$$

$$x^2 = (e^y + 2)^2$$

$$= e^{2y} + 4e^y + 4$$

$$\begin{aligned}
 V &= \pi \int_0^1 (e^{2y} + 4e^y + 4) dy \\
 &= \pi \left[\frac{1}{2}e^{2y} + 4e^y + 4y \right]_0^1 \\
 &= \pi \left(\frac{1}{2}e^{2(1)} + 4e^{(1)} + 4(1) \right) \\
 &\quad - \pi \left(\frac{1}{2}e^{2(0)} + 4e^{(0)} + 4(0) \right) \\
 &= \pi \left(\frac{1}{2}e^2 + 4e - \frac{1}{2} \right) \text{ units}^3
 \end{aligned}$$

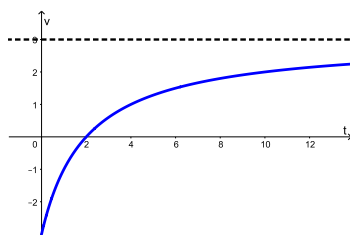
Question 15 :

- (a) i. (1 mark) Stationary when $v = 0$

$$\begin{aligned}
 3 - \frac{12}{2+t} &= 0 \\
 \frac{12}{2+t} &= 3 \\
 3(2+t) &= 12 \\
 2+t &= 4 \\
 t &= 2 \text{ seconds}
 \end{aligned}$$

- ii. (1 mark) As $t \rightarrow \infty, v \rightarrow 3 \text{ m/s}$

- iii. (1 mark)



- iv. (2 marks)

$$\begin{aligned}
 a &= \frac{d}{dt} \left(3 - \frac{12}{2+t} \right) \\
 &= \frac{d}{dt} (3 - 12(2+t)^{-1}) \\
 &= 12(2+t)^{-2} \\
 &= \frac{12}{(2+t)^2}
 \end{aligned}$$

$$\begin{aligned}
 a(2) &= \frac{12}{(2+2)^2} \\
 &= \frac{3}{4} \text{ m/s}^2
 \end{aligned}$$

- v. (3 marks)

$$x(t) = 3t - 12 \ln(t+2) + C$$

$$\begin{aligned}
 x(0) &= 3(0) - 12 \ln(0+2) + C \\
 &= -12 \ln 2 + C
 \end{aligned}$$

$$\begin{aligned}
 x(2) &= 3(2) - 12 \ln(2+2) + C \\
 &= 6 - 12 \ln 4 + C \\
 &= 6 - 24 \ln 2 + C
 \end{aligned}$$

$$\begin{aligned}
 x(6) &= 3(6) - 12 \ln(6+2) + C \\
 &= 18 - 12 \ln 8 + C \\
 &= 18 - 36 \ln 2 + C
 \end{aligned}$$

particle moves left $t = 0$ to $t = 2$:
distance = $x(0) - x(2)$
 $= (-12 \ln 2 + C) - (6 - 24 \ln 2 + C)$
 $= 12 \ln 2 - 6$

particle moves right $t = 2$ to $t = 6$:
distance = $x(6) - x(2)$
 $= (18 - 36 \ln 2 + C) - (6 - 24 \ln 2 + C)$
 $= 12 - 12 \ln 2$

Total distance = $(12 \ln 2 - 6) + (12 - 12 \ln 2) = 6$ metres

- (b) i. (2 marks) $\frac{d}{dx} (x^2 e^{-x^2})$:
Let $u = x^2, u' = 2x$
Let $v = e^{-x^2}, v' = -2x e^{-x^2}$

$$\begin{aligned}
 \text{LHS} &= \frac{d}{dx} (x^2 e^{-x^2}) \\
 &= u'v + uv' \\
 &= 2x \times e^{-x^2} - x^2 \times -2x e^{-x^2} \\
 &= 2x e^{-x^2} - 2x^3 e^{-x^2} \\
 &= \text{RHS}
 \end{aligned}$$

- ii. (2 marks) From (i),

(b) i. (1 mark)

 AP has gradient 1, so $NP = AN$.From the diagram, $ON = 1 + AN$ $\therefore ON = 1 + NP$ $\therefore NP = ON - 1$

$$\frac{d}{dx} (x^2 e^{-x^2}) = 2x^3 e^{-x^2} - 2x e^{-x^2}$$

$$2 \int x^3 e^{-x^2} dx = \int 2x e^{-x^2} dx - \int \left(\frac{d}{dx} x^2 e^{-x^2} \right) dx$$

$$2 \int x^3 e^{-x^2} dx = \int 2x e^{-x^2} dx - x^2 e^{-x^2}$$

$$2 \int x^3 e^{-x^2} dx = -e^{-x^2} - x^2 e^{-x^2} + C \text{ from (i)}$$

$$\int x^3 e^{-x^2} dx = -\frac{1}{2} (e^{-x^2} + x^2 e^{-x^2}) + C$$

ii. (1 mark) OP has gradient x ,

$$\therefore \frac{NP}{ON} = x$$

$$\therefore NP = xON$$

iii. (1 mark)

$$ON - 1 = xON \text{ from (i), (ii)}$$

$$ON - xON = 1$$

$$ON(1 - x) = 1$$

$$ON = \frac{1}{1 - x}$$

From the diagram,

$$ON = 1 + x + x^2 + x^3 + \dots$$

$$\therefore 1 + x + x^2 + x^3 + \dots = \frac{1}{1 - x}$$

iv. If $x \geq 1$ then the line OB will not intersect the gradient=1 line AP .

or

If $x \geq 1$ then $x^2 > x$, $x^3 > x^2$, etc, so the sequence of triangles will not reduce in size, so the diagram will not work.

NB: the distances will not become negative if $x \geq 1$ - it is possible to draw this condition, however the lines OB and AP do not intersect.

(c) i. (1 mark)

$\triangle HJK$ is an equilateral triangle,
sides 2 units

$$\text{area} = \frac{1}{2} \times 2 \times 2 \times \sin \pi/3$$

$$= \sqrt{3}u^2$$

ii. (2 marks)

$$\text{segment area} = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= \frac{2\pi - 3\sqrt{3}}{12}$$

$$\therefore \text{Area} = \sqrt{3} + 6 \times \frac{2\pi - 3\sqrt{3}}{12}$$

$$= \left(\pi - \frac{\sqrt{3}}{2} \right) u^2$$

Question 16 :

(a) (2 marks)

$$\text{LHS} = a^{\log_b x}$$

$$= \left(x^{\log_x a} \right)^{\log_b x}$$

$$= x^{\log_x a \times \log_b x}$$

$$= x^{\frac{\ln a}{\ln x} \times \frac{\ln x}{\ln b}}$$

$$= x^{\frac{\ln a}{\ln b}}$$

$$= x^{\log_b a}$$

$$= \text{RHS}$$

(c) i. (1 mark)

 $K'B$ is $D \tan \theta$

$$\text{S travel time} = \frac{D \tan \theta}{V}$$

Sandra takes 1 hr to reach K' , so

$$t = \frac{D \tan \theta}{V} + 1$$

ii. (2 marks)

 KB is $D \sec \theta$

Keanu needs to travel this distance

in the time S reaches B

$$\begin{aligned} r &= (D \sec \theta) \div \left(\frac{D \tan \theta}{V} + 1 \right) \\ &= (D \sec \theta) \div \left(\frac{D \tan \theta + V}{V} \right) \\ &= \frac{DV \sec \theta}{D \tan \theta + V} \end{aligned}$$

iii. (2 marks)

$$\frac{dr}{d\theta} = \frac{u'v - uv'}{v^2}$$

$$\begin{aligned} uv' - uv' &= (DV \sec \theta \tan \theta)(D \tan \theta + V) \\ &\quad - (D \sec^2 \theta)(DV \sec \theta) \\ &= DV \sec \theta (D \tan^2 \theta \\ &\quad + V \tan \theta - D \sec^2 \theta) \\ &= DV \sec \theta (D \tan^2 \theta + V \tan \theta \\ &\quad - D(1 + \tan^2 \theta)) \\ &= DV \sec \theta (V \tan \theta - D) \end{aligned}$$

$$\frac{dr}{d\theta} = \frac{DV \sec \theta (V \tan \theta - D)}{(D \tan \theta + V)^2}$$

iv. (2 marks)

Stationary points when $\frac{dr}{d\theta} = 0$.

$\sec \theta > 0$ for the domain $0 \leq \theta < \frac{\pi}{2}$.

$(V \tan \theta - D) = 0$ when $\tan \theta = \frac{D}{V}$.

Let θ_m be the value for which $\tan \theta_m = \frac{D}{V}$

To show that r is a minimum for θ_m , we must test using the first or second derivative (!).

Consider $\alpha < \theta_m$.

Since $\tan \theta$ is an increasing function

for $0 \leq \theta < \frac{\pi}{2}$,

$$\tan \alpha < \tan \theta_m < \frac{D}{V}$$

$$\therefore (V \tan \alpha - D) < (V \frac{D}{V} - D) < 0$$

Consider $\beta > \theta_m$.

Since $\tan \theta$ is an increasing function for $0 \leq \theta < \frac{\pi}{2}$,

$$\tan \beta > \tan \theta_m > \frac{D}{V}$$

$$\therefore (V \tan \beta - D) > (V \frac{D}{V} - D) > 0$$

θ	α	θ_m	β
$\sec \theta - D$	(+)	(+)	(+)
$V \tan \theta - D$	(-)	0	(+)
$(D \tan \theta + V)^2$	(+)	(+)	(+)
$\frac{dr}{d\theta}$	(+)(-)/(+)	0	(+)(+)/(+)
	\	—	/

Therefore r is a minimum when $\tan \theta = \frac{D}{V}$.

v. (2 marks)

If $\tan \theta = \frac{D}{V}$, draw diagram

then $\sec \theta = \frac{\sqrt{D^2 + V^2}}{V}$, (θ acute).

$$\begin{aligned} r &= \frac{DV \sec \theta}{D \tan \theta + V} \\ &= \frac{DV \frac{\sqrt{D^2 + V^2}}{V}}{D(\frac{D}{V}) + V} \\ &= \frac{D(\sqrt{D^2 + V^2})}{\frac{D^2}{V} + V} \\ &= \frac{D(\sqrt{D^2 + V^2})}{\frac{D^2 + V^2}{V}} \\ &= \frac{DV(\sqrt{D^2 + V^2})}{D^2 + V^2} \\ &= \frac{DV}{\sqrt{D^2 + V^2}} \end{aligned}$$